

Shailza Jolly

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📍 Berlin, Germany

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Summary

- ML Scientist/Engineer with a Ph.D. in CS and 6+ years spanning academic research and industry, working at the intersection of science, engineering, and product. Specializes in LLMs, Generative AI, and scalable data pipelines, with hands-on depth in modern LLM architectures, efficiency techniques (KV-cache, speculative decoding, FlashAttention), and agentic system design. Pairs a track record of top-tier publications (AAAI, NAACL, EMNLP) with shipping production systems and translating technical work into clear business outcomes.

Experience

Independent, Applied AI & LLM Systems

Berlin, Germany

July 2025 – present

1 year

- Worked hands-on at the intersection of science, engineering, and product, building and iterating on a series of prototypes across agentic systems, retrieval, and LLM evaluation to keep my engineering sharp while exploring directions worth committing to.
- Refreshed modern LLM foundations (Stanford) and went deep on architecture and efficiency internals (KV-cache, speculative decoding, FlashAttention). One example is [Sift](#), an agentic RAG system that returns grounded, citable answers over a corpus of technical PDFs.

Flinn.ai, Senior AI Engineer

Berlin, Germany

Jan 2025 – June 2025

6 months

- Led development of AI-driven product features including data extraction from medical research papers and multilingual complaint monitoring.
- Partnered with product and backend teams to scope problems, define success metrics, and deliver end-to-end features within an agile product lifecycle.

Parental Leave (Maternity)

Berlin, Germany

Dec 2023 – Dec 2024

1 year 1 month

- Stayed engaged with research literature and maintained technical skills.

Amazon AI, Research Scientist

Berlin, Germany

July 2022 – Nov 2023

1 year 5 months

- Led development of a scalable noise-removal/data-quality pipeline processing billions of tokens to improve training data for LLMs.
- Wrote research plans and technical documentation; presented results and recommendations to senior science/engineering leadership.
- Mentored Master's/Ph.D. interns and collaborated on research deliverables.

German Research Center for Artificial Intelligence (DFKI), Machine Learning Scientist

Berlin, Germany

Mar 2019 – June 2022

3 years 4 months

- Delivered ML research and engineering in BMBF-funded projects XAINES (Explainable AI) and DeFuseNN (vision-language systems).
- Supervised interns and BSc/MSc students; provided technical mentorship and project guidance.

NVIDIA Research, Machine Learning Scientist Intern

Santa Clara, US

May 2021 – Aug 2021

4 months

- Built a document understanding pipeline combining table/cell detection, tabular structure retrieval, and OCR for financial documents.

Amazon Alexa, Applied Scientist Intern

Aachen, Germany

Aug 2019 – Dec 2019

5 months

- Developed a method for generating diverse synthetic training data to improve intent classification and slot labeling for task-oriented NLU.

- Designed and implemented an evaluation metric for Visual Question Answering (VQA) models.

Skills

ML/GenAI: LLMs, NLU/NLG, multimodal learning, synthetic data, model evaluation

LLM Systems: Agentic workflows (MCP, LangGraph/LangSmith), RAG, grounding, experiment design, offline evaluation

Frameworks & Libraries: Python, PyTorch, Hugging Face (Transformers, PEFT, Datasets), PySpark, Weights & Biases

Search & Retrieval: Sentence Transformers, reranking, vector search, Pinecone, Chroma

Infrastructure & Deployment: AWS, Docker, FastAPI, Git

Education

Ph.D. TU Kaiserslautern , Computer Science	Kaiserslautern, Germany
- Grade: Sehr Gut 1.0 (highest on 1.0–5.0 scale) - Thesis: Building Natural Language Generation and Understanding Systems in Data-Constrained Settings - Work on VQA, interpretability, and conversational AI	2019 – 2022
Visiting University of Copenhagen , Natural Language Processing	Copenhagen, Denmark
Developed an unsupervised post-editing algorithm to generate fluent fact-check-Research explanations	Jan 2021 – Mar 2021
M.Sc. TU Kaiserslautern , Computer Science — Minor in Economics	Kaiserslautern, Germany
Grade: Sehr Gut 1.5	2016 – 2018
Seimes- Kyushu University , Computer Science	Fukuoka, Japan
Explainable AI to analyze behavior of deep CNN architectures for image recognition	Nov 2017 – Feb 2018
B.Tech Guru Nanak Dev Engineering College , Computer Science & Engineering	India
Grade: First Division with Distinction	2012 – 2016

Selected Publications

Search and Learn: Improving Semantic Coverage for Data-to-Text Generation	Jan 2022
Shailza Jolly, Z. X. Zhang, A. Dengel, L. Mou arxiv.org/abs/2112.02770 (AAAI)	
EaSe: A Diagnostic Tool for VQA Based on Answer Diversity	Jan 2021
Shailza Jolly, S. Pezzelle, M. Nabi www.aclweb.org/anthology/2021.naacl-main.192 (NAACL-HLT)	
Can Pre-training Help VQA with Lexical Variations?	Jan 2020
Shailza Jolly, S. Kapoor www.aclweb.org/anthology/2020.findings-emnlp.257 (Findings of EMNLP)	
Data-Efficient Paraphrase Generation to Bootstrap Intent Classification and Slot Labeling for New Features in Task-Oriented Dialog Systems	Jan 2020
Shailza Jolly, T. Falke, C. Tirkaz, D. Sorokin www.aclweb.org/anthology/2020.coling-industry.2 (COLING (Industry Track))	
How Do Convolutional Neural Networks Learn Design?	Jan 2018
Shailza Jolly, B. K. Iwana, R. Kuroki, S. Uchida ieeexplore.ieee.org/abstract/document/8545624 (International Conference on Pattern Recognition (ICPR))	

Awards

AAAI-22 Scholarship by Hitachi

Jan 2022

- Awarded by Hitachi to attend AAAI 2022.

AI Newcomer 2021 Award

Jan 2021

- Recognized by the German Informatics Society and BMBF, Germany.

EU-Cost STSM Grant

Jan 2020

- Awarded by EU-Cost Action to work on the Multi3generation project at CopeNLU, University of Copenhagen.

Best Student Paper Award

Jan 2018

- Awarded by the International Conference on Pattern Recognition (ICPR) for "How Do Convolutional Neural Networks Learn Design?"

Media Coverage

- [Interview featured in the Rheinfalz newspaper.](#)
- Radio interview for Antenne Kaiserslautern, Germany.

Languages

English: Fluent

German: Beginner

Hindi: Native speaker